

Week01.01

GEOG 4/5/7 9073: Environmental Analysis in R:

Dr. Patrick Bitterman

2026-01-13

Today's schedule

- Open discussion
- Course welcome
- Introductions
- R basics and practice

Anything to discuss? Questions?

My approach to this course

- Geography matters
- Concepts and independent thinking are important, trivia is not (or at least not always)
- It's important to solve problems or complete tasks, but understanding HOW you do so is more important

What this class is

- Collaborative
- Flexible
- *Student-led*

What else it is

- A bit mis-named
- My second time teaching it here at KSU - so we have space to experiment a bit

Class introductions

Pair up – and don't leave anyone behind

Share:

- Name
- Where you're from
- Major/program, year
- Why are you taking this course?
- Why are you in your major? Why KSU?
- Previous experience: GIS, programming, R?
- Imagine it's May 2026 – what would make you feel like you were successful in this course. SAVE THIS SOMEWHERE!!!

My introduction

- Dr. Patrick Bitterman
- Independence, IA -> Ulowa -> UVM -> UNL -> KSU
- Assistant Professor in Geography Department, 2nd/7th year
- Goals: teach a successful course, build my research program
- GIS experience? Programming experience? Extensive, but I don't use ArcGIS much...
- Success in this course:
 - students reach *their* learning objectives
 - students are able to use R to make their work faster and more consistent
 - students improve methods related to their other research or career interests
 - students develop an appreciation for programmatic spatial analysis

Time to share

- Present your partner to the class
- Who wants to go first?

Share:

- Name
- Where you're from
- Major/program, year
- Why are you taking this course?
- Why are you in your major? Why KSU?
- Previous experience: GIS, programming, R?
- What would make you feel successful?

Course basics

Instructor (me)

- Dr. Patrick Bitterman
- Geography Department
- Office: 436 McGilvrey Hall
- Office hours: MW 9-11am, or by appointment
- Email: pbitterm@kent.edu

Materials

- Brunsdon, C., Comber, L. 2025. An Introduction to R for Spatial Analysis and Mapping (Spatial Analytics and GIS). 3rd Edition. SAGE. [Link](#)
- Optional: Wickham, H. 2017. R for Data Science: Import, Tidy, Transform, Visualize, and Model Data. O'Reilly Media. (if you are unfamiliar with R or other programming language) [Link](#)

Learning objectives

By the end of the term, students will be able to successfully:

- Demonstrate a familiarity with the R programming language in the context of geospatial analysis
- Write self-contained functions to automate geospatial tasks
- Analyze model workflows and describe computer code and algorithms in plain language
- Create small-scale programs that interface with web-based tools
- Practice good programming practices
- Plan, develop, and execute a programmatic analysis of a dataset

Course policies

- Assignment submission:
 - All assignments due on their due date
 - All assignments will be posted on Canvas, but turned in via GitHub (we'll talk more)
 - Late items will be accepted, but will be penalized 20% of the potential points for each day they are late
- All changes to the syllabus will be communicated via Canvas announcement
- Students are expected to attend all class meetings, but attendance is not graded. **Participation is graded, though**

Collaboration

- Feel free to discuss labs, etc. with your classmates
- However...
 - All lab reports, papers, and other work should be your own, individual thoughts
 - Students who do not follow these policies will be reported to the College for academic dishonesty

Other tips

- Read relevant materials before class
- Attend class – understanding theory and concepts will help you with practical applications
- If there are topics, news stories, blog posts, tweets, etc. that you find interesting or want to know more about, let me know
- Before you start coding, think through the process and sketch out the workflow. This is called *pseudocode*
- Labs build on each other, so don't get behind
- Take advantage of office hours
- Do not leave assignments until the last minute
- Have fun!

Assessment

Lab assignments

- 6 labs, 2-3 weeks to complete each one

Final project

- Proposal
- Update presentation (in-class)
- Final presentation (in-class)
- Final report

Participation

Graduate students

For graduate students, the requirements of the final project will be expanded to include:

1. an additional 3-4 pages in your report
2. code documentation
3. an additional 5 minutes in your final presentation to the class.
4. required to produce a cover page for your GitHub page/portfolio

Course format

- Project-based
- Student-led
 - I am not going to recap the readings (much)
 - You are expected to be ready to participate in discussion
 - We will spend most of our time doing, not lecturing
- **Tuesdays:** discussion, examples, and activities
- **Thursdays:** wildcard day (coding challenges, group work, seminar)

Course inspiration and source material

- My own experience
- Your textbook
- Lovelace, R. *Geocomputation with R*
<https://geocompr.robinlovelace.net/index.html>

University policies

- Learning accommodations
 - Contact Student Support Services
 - Let me know
- Academic integrity
 - Don't cheat
 - Don't plagiarize
- Health and safety
 - Student support services (<https://www.kent.edu/studentsupportservices>){.uri}

The Generative AI Elephant in the Room

- Yes, ChatGPT, Claude, Copilot, etc. can write code...
- ...and most of the time, it's even correct

For example...

write some example R code that shows how to join two data frames together

Here's R code demonstrating the main approaches for joining data frames:

Claude output

```
library(tidyverse)
# Create example data frames
df1 <- data.frame(
  studentId = c(1, 2, 3, 4),
  name = c("Alice", "Bob", "Carol", "David"),
  score = c(85, 92, 78, 88)
)

df2 <- data.frame(
  studentId = c(2, 3, 4, 5),
  department = c("Biology", "Chemistry", "Physics", "Math"),
  year = c(2, 3, 1, 4)
)

inner_join(df1, df2, by = "studentId")
```

```
##   studentId  name score department year
## 1         2   Bob   92    Biology    2
## 2         3  Carol   78    Chemistry  3
## 3         4  David   88    Physics    1
```

The AI “discussion”

In your opinion AND in the context of this class...

- What are some arguments FOR using GAI for writing code?
- What are some arguments AGAINST using GAI for writing code?
- What is a reasonable use policy?

Anything else?

- Questions?

Let's look at the Canvas and GitHub sites

Let's talk writing code

SA and Programming in R

- Experience? In what setting?
- Experience with a version control system (e.g., git)?
- What about experience with spatial analysis, GIS?

Let's make sure we're setup and familiar with R + RStudio

1. Login to your computer if you haven't (you may also use your own)
2. Launch RStudio
3. Wait

R and RStudio: Understanding the Relationship

R and RStudio are complementary but distinct tools that work together for data analysis:

R

- underlying statistical programming language and computational engine
- performs all calculations, executes code, manages data structures, and produces outputs
- can run independently from the command line or terminal

RStudio

- is an Integrated Development Environment (IDE)
- sends your code to R for execution and displays the results in an organized workspace
- means you could use R with different interfaces (VSCode, Jupyter, command line), but...
- RStudio is purpose-built for R workflows and offers features specifically designed for statistical analysis and data science tasks

The RStudio Interface

When you first launch RStudio, you'll see a workspace divided into four main panes (which you can customize):

1. Source Pane (Top Left)

This is your code editor where you write and save R scripts (.R files) or Quarto/R Markdown documents (.qmd/.Rmd files).

- Write multi-line code that you can save and reuse
- Execute single lines with **Ctrl+Enter** (Windows/Linux) or **Cmd+Enter** (Mac)
- Execute selected code chunks
- Syntax highlighting helps identify functions, variables, and errors
- Line numbers assist with debugging

Always (mostly) work in scripts rather than only in the Console—this ensures reproducibility.

2. Console Pane (Bottom Left)

This is where R actually runs and displays output.

- Code executed from the Source pane appears here
- You can type code directly for quick tests or calculations
- Results, messages, warnings, and errors display here
- The `>` prompt indicates R is ready for input
- The `+` prompt means R is waiting for you to complete a command

3. Environment/History Pane (Top Right)

Environment tab:

- Shows all objects currently loaded in your R session (e.g., data frames, vectors, functions)
- Click on data frames to view them in a spreadsheet-like format
- Displays object types and dimensions
- The broom icon clears your workspace (be careful!)

History tab:

- Records all commands you've executed
- Useful for recovering code you ran but didn't save

4. Files/Plots/Packages/Help Pane (Bottom Right)

multi-purpose pane with tabs:

- Files
- Plots
- Packages
- Help
- Viewer

Let's try it out

Practice using RStudio by:

1. Creating a new R Script (File > New File > R Script)
2. Writing simple code (e.g., `x <- 1:10`)
3. Executing it and observing results in the Console
4. Viewing created objects in the Environment pane
5. Creating a simple plot and viewing it in the Plots pane

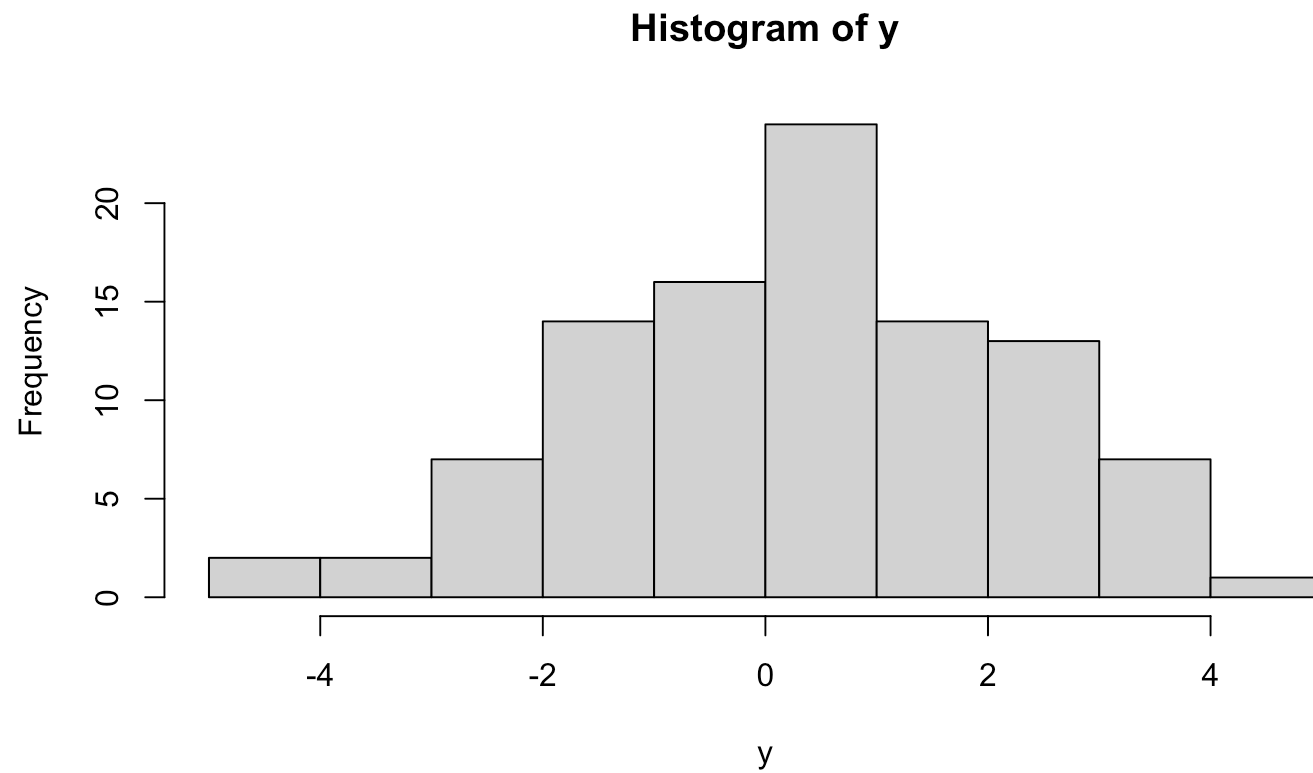
```
x <- 1:10  
x
```

```
## [1] 1 2 3 4 5 6 7 8 9 10
```

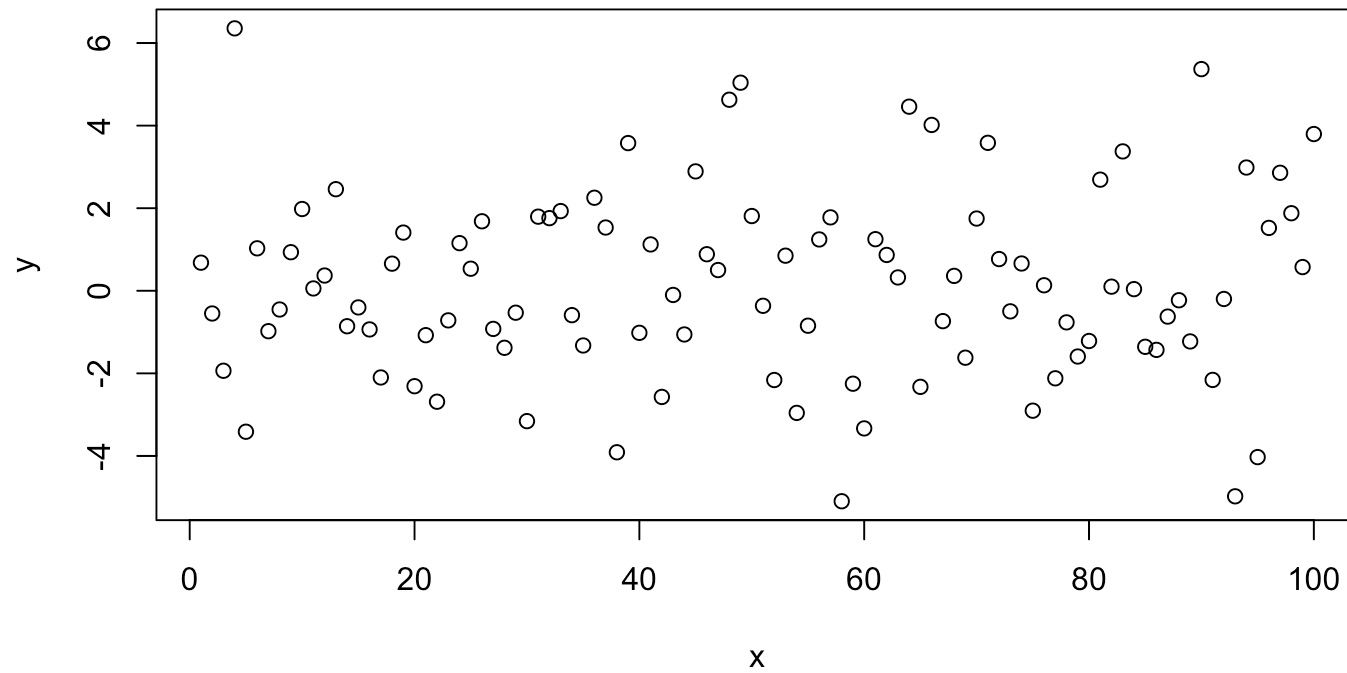
```
x * 2
```

```
## [1] 2 4 6 8 10 12 14 16 18 20
```

```
y <- rnorm(100, mean = 0, sd = 2)  
hist(y)
```



```
x <- 1:100  
y <- rnorm(100, mean = 0, sd = 2)  
plot(x, y)
```



Review and next class

- Any questions on course policies?
- On anything else?

This week's readings/tasks:

1. Complete the setup steps in the [“getting up to speed”](#) page
2. Review chapters 1 - 4 on the *R for Data Science* page I linked above. Don't worry, it'll go more quickly than it sounds.
3. Practice on your own
4. Come to **NEXT** class with **at least 2 questions** you have about R, geospatial programming in general, or this course.

Next session: basics of R and GIScience